Experiment 1 — Data Preprocessing

Part 1 — Python Code

```
# Step 2: Import the required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler, MinMaxScaler

# Step 1 & 3: Load the dataset and inspect structure
df = pd.read_csv("https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv")

print("First 5 Rows:")
print(df.head())

print("\nDataset Info:")
df.info()

print("\nSummary Statistics:")
print(df.describe())

# Step 4: Identify and handle missing values
print("\nMissing values per column:")
print(df.isnull().sum())

# Fill numerical missing values with median
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Fare'] = df['Fare'].fillna(df['Fare'].median())

# Fill categorical missing values with mode
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# Drop column with excessive missing data
if 'Cabin' in df.columns:
    df.drop(columns=['Cabin'], inplace=True)

# Step 5: Detect and remove duplicate records
print(f"\nDuplicates found: {df.duplicated().sum()}")
df.drop_duplicates(inplace=True)

# Step 6: Encode categorical variables
# Label Encoding for binary features
le = LabelEncoder()
df['Sex'] = le.fit_transform(df['Sex'])  # male: 1, female: 0

# One-Hot Encoding for multi-class categorical features
df = pd.get_dummies(df, columns=['Embarked'], drop_first=True)

# Step 7: Outlier Detection and Treatment (IQR Method for Fare)
Q1 = df['Fare'].quantile(0.25)
Q3 = df['Fare'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Cap outliers
df['Fare'] = np.clip(df['Fare'], lower_bound, upper_bound)

# Step 8: Normalization / Standardization
scaler = StandardScaler()
df[['Age', 'Fare']] = scaler.fit_transform(df[['Age', 'Fare']])

# Step 9: Feature Engineering
# Create FamilySize = SibSp + Parch + 1 (self)
df['FamilySize'] = df['SibSp'] + df['Parch'] + 1

# Create AgeGroup category
df['AgeGroup'] = pd.cut(df['Age'], bins=3, labels=['Child', 'Adult', 'Senior'])

# Step 10: Save cleaned dataset
df.to_csv('titanic_cleaned.csv', index=False)
print("\nPreprocessing complete. Saved to 'titanic_cleaned.csv'.")
```

Part 2 — Questions & Answers

1. Why is data preprocessing considered one of the most important phases in data analytics?

Solution:

Raw data is often noisy, missing values, duplicated, and inconsistent. Preprocessing cleans and structures data so machine learning algorithms can converge efficiently and produce accurate, unbiased predictions without "garbage in, garbage out" failure.

2. Explain different methods of handling missing values with suitable examples.

Solution:

Deletion: Dropping rows (df.dropna()) when missing entries are rare (<5%), or dropping columns (df.drop()) if most entries are missing (e.g., Cabin column).

Mean/Median Imputation: Replacing missing numbers with the central tendency (e.g., filling missing Age with median 28 to avoid skew).

Mode Imputation: Replacing missing categorical values with the most frequent value (e.g., filling missing Embarked with 'S').

Predictive/KNN Imputation: Estimating missing values based on similarities to other feature rows.

3. Differentiate between Label Encoding and One-Hot Encoding.

Solution:

FeatureLabel EncodingOne-Hot EncodingMechanismAssigns an integer to each category ($0, 1, 2...$)Creates separate binary columns (0 or 1) for each unique valueBest Used ForOrdinal variables with inherent order (e.g., Low, Medium, High)Nominal variables without ordering (e.g., City: Delhi, Mumbai, Pune)RiskModels may misinterpret numerical order as mathematical weightIncreases dimensionality significantly (Curse of Dimensionality)

4. What are outliers? How can they affect analytical results?

Solution:

Outliers are data points that deviate significantly from the overall pattern of the data. They skew summary statistics (inflating the mean and standard deviation) and degrade the performance of sensitive models (such as Linear Regression, K-Means clustering, and Neural Networks).

5. Explain the difference between normalization and standardization.

Solution:

Normalization (Min-Max Scaling): Rescales values to a fixed range, usually $[0, 1]$, using:$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$Useful for algorithms requiring bounded inputs (e.g., Neural Networks, KNN).Standardization (Z-Score Normalization): Centers data around a mean of 0 with a standard deviation of 1 using:$$Z = \frac{X - \mu}{\sigma}$$Less sensitive to outliers and standard for algorithms assuming Gaussian distributions (e.g., Logistic Regression, SVM, PCA).

6. Why should duplicate records be removed before analysis?

Solution:

Duplicate records lead to data leakage, falsely amplify patterns, skew model weights toward overrepresented points, and result in overoptimistic/inaccurate evaluation metrics.

7. What is feature engineering? Give two practical examples.

Solution:

Feature engineering is the process of using domain knowledge to extract new variables from raw data to improve model predictive power.Example 1: Combining SibSp (siblings/spouses) and Parch (parents/children) into a single FamilySize metric.Example 2: Extracting DayOfWeek, Hour, or IsWeekend from a single Timestamp column.

8. Which preprocessing techniques would you apply to the IBM HR Employee Attrition dataset and why?

Solution:

Removal of Zero-Variance Columns: Drop columns with a single unique value (like EmployeeCount, StandardHours, Over18) as they provide zero information.ID Removal: Drop identifier columns (EmployeeNumber) to prevent overfitting.One-Hot / Binary Encoding: Convert categorical attributes (OverTime, Department, JobRole, BusinessTravel) to numeric vectors.Feature Scaling: Apply StandardScaler to features like MonthlyIncome, TotalWorkingYears, and Age because they exist on vastly different numerical scales.Class Imbalance Handling: Handle the imbalanced target class (Attrition yes/no) using techniques like SMOTE or class weighting.

9. How does poor-quality data affect machine learning model performance?

Solution:

It causes biased estimates, high variance, poor generalizability on unseen data, slow convergence during training, and misleading feature importance rankings.

10. Name any three Python libraries commonly used for data preprocessing.

Solution:

Pandas (Data manipulation and cleaning)NumPy (Numerical operations and array handling)Scikit-Learn (sklearn.preprocessing module for scaling, encoding, and imputation)

```
First 5 Rows:
   PassengerId  Survived  Pclass  ...     Fare Cabin  Embarked
0            1         0       3  ...   7.2500   NaN         S
1            2         1       1  ...  71.2833   C85         C
2            3         1       3  ...   7.9250   NaN         S
3            4         1       1  ...  53.1000  C123         S
4            5         0       3  ...   8.0500   NaN         S

[5 rows x 12 columns]

Dataset Info:
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB

Summary Statistics:
       PassengerId    Survived      Pclass         Age       SibSp       Parch        Fare
count   891.000000  891.000000  891.000000  714.000000  891.000000  891.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008    0.381594   32.204208
std     257.353842    0.486592    0.836071   14.526497    1.102743    0.806057   49.693429
```